

```

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy import stats

# -----
# Sample Data
# -----
X = np.array([2, 4, 6, 8, 10])
Y = np.array([1, 3, 5, 7, 9])

# -----
# Basic Statistics
# -----
mean_x = np.mean(X)
variance_x = np.var(X)          # population variance
std_dev_x = np.std(X)           # population standard deviation

print("Mean:", mean_x)
print("Variance:", variance_x)
print("Standard Deviation:", std_dev_x)

# -----
# Correlation
# -----
correlation = np.corrcoef(X, Y)[0, 1]
print("Correlation between X and Y:", correlation)

# -----
# Normal Distribution
# -----
normal_data = np.random.normal(loc=mean_x, scale=std_dev_x, size=1000)

# -----
# Uniform Distribution
# -----
uniform_data = np.random.uniform(low=0, high=10, size=1000)

# -----
# Plot Distributions
# -----
plt.figure(figsize=(10, 4))

plt.subplot(1, 2, 1)
plt.hist(normal_data, bins=30, density=True)
plt.title("Normal Distribution")

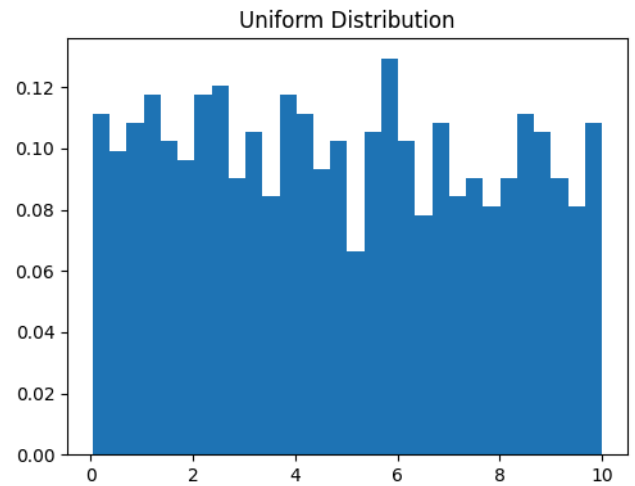
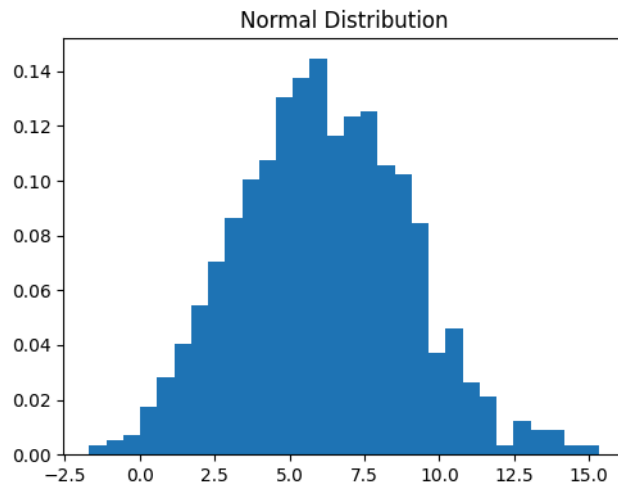
plt.subplot(1, 2, 2)
plt.hist(uniform_data, bins=30, density=True)
plt.title("Uniform Distribution")

plt.tight_layout()
plt.show()

# -----
# Using Pandas (optional)
# -----
df = pd.DataFrame({"X": X, "Y": Y})
print("\nPandas Summary:")
print(df.describe())
print("\nPandas Correlation:")
print(df.corr())

```

Mean: 6.0
 Variance: 8.0
 Standard Deviation: 2.8284271247461903
 Correlation between X and Y: 0.9999999999999999



Pandas Summary:

	X	Y
count	5.000000	5.000000
mean	6.000000	5.000000
std	3.162278	3.162278
min	2.000000	1.000000
25%	4.000000	3.000000
50%	6.000000	5.000000
75%	8.000000	7.000000
max	10.000000	9.000000

Pandas Correlation:

	X	Y
X	1.0	1.0
Y	1.0	1.0

```
import numpy as np
import matplotlib.pyplot as plt

# -----
# Sample Data
# -----
data1 = np.array([10, 12, 15, 18, 20, 22, 25, 28, 30, 35])
data2 = np.array([8, 11, 14, 17, 19, 21, 24, 27, 29, 34])

# -----
# Mean
# -----
mean_value = np.mean(data1)
print("Mean:", mean_value)

# -----
# Variance
# -----
variance_value = np.var(data1)
print("Variance:", variance_value)

# -----
# Standard Deviation
# -----
std_dev = np.std(data1)
print("Standard Deviation:", std_dev)

# -----
# Correlation
# -----
correlation = np.corrcoef(data1, data2)
print("Correlation Matrix:\n", correlation)

# -----
# Normal Distribution
# -----
normal_data = np.random.normal(mean_value, std_dev, 1000)

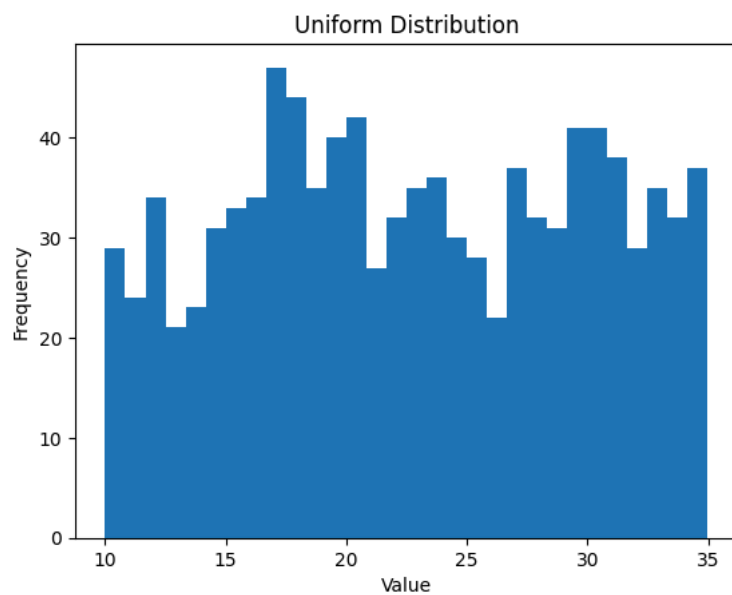
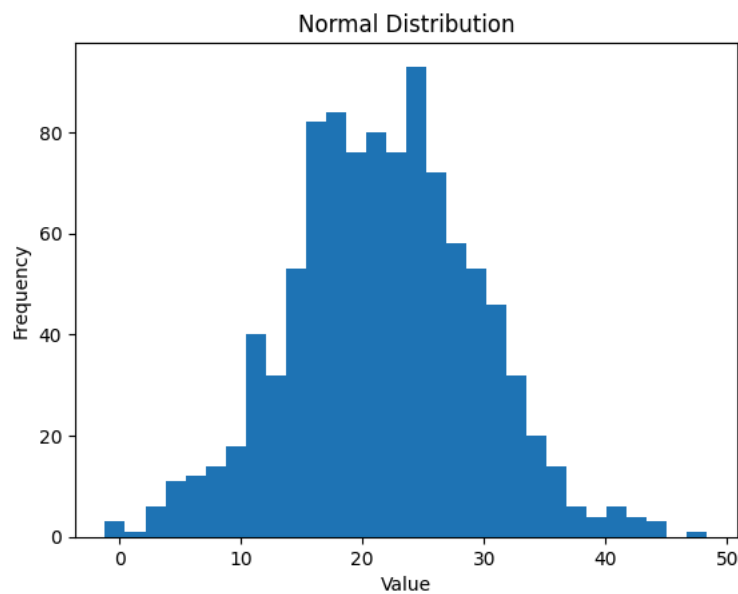
# -----
# Uniform Distribution
# -----
```

```
uniform_data = np.random.uniform(min(data1), max(data1), 1000)
```

```
# -----
# Plot Distributions
# -----
plt.figure()
plt.hist(normal_data, bins=30)
plt.title("Normal Distribution")
plt.xlabel("Value")
plt.ylabel("Frequency")
plt.show()

plt.figure()
plt.hist(uniform_data, bins=30)
plt.title("Uniform Distribution")
plt.xlabel("Value")
plt.ylabel("Frequency")
plt.show()
```

```
Mean: 21.5
Variance: 58.85
Standard Deviation: 7.671375365604267
Correlation Matrix:
[[1.          0.99944851]
 [0.99944851  1.        ]]
```



```
import pandas as pd
import numpy as np

# -----
# LOAD DATASET USING FILE PATH
# -----
```

```
# For CSV file
df = pd.read_csv('/content/Housing - Housing.csv')

# If your file is Excel, use:
# df = pd.read_excel(path)

print("\nOriginal Dataset:")
print(df)

# -----
# 1. Missing Value Analysis
# -----
print("\nMissing Values Count:")
print(df.isnull().sum())

# -----
# 2. Fixing Missing Values (Mean Imputation)
# -----
df_filled = df.fillna(df.mean(numeric_only=True))

print("\nDataset After Fixing Missing Values:")
print(df_filled)

# -----
# 3. Outlier Analysis using IQR
# -----
Q1 = df_filled.quantile(0.25, numeric_only=True)
Q3 = df_filled.quantile(0.75, numeric_only=True)
IQR = Q3 - Q1

outliers = (df_filled < (Q1 - 1.5 * IQR)) | (df_filled > (Q3 + 1.5 * IQR))

print("\nOutlier Detection (True = Outlier):")
print(outliers)

# Remove outliers
df_cleaned = df_filled[~outliers.any(axis=1)]

print("\nDataset After Removing Outliers:")
print(df_cleaned)
```



```

import pandas as pd
import numpy as np

# -----
# MANUALLY GIVEN DATASET
# -----
# You can change values as you want
Age = [20, 22, 25, None, 28, 30, 100]
Salary = [25000, 27000, None, 30000, 29000, 32000, 100000]
Experience = [1, 2, 3, 2, None, 4, 30]

# Create DataFrame
df = pd.DataFrame({
    'Age': Age,
    'Salary': Salary,
    'Experience': Experience
})

print("Original Dataset:")
print(df)

# -----
# 1. Missing Value Analysis
# -----
print("\nMissing Values Count:")
print(df.isnull().sum())

# -----
# 2. Fixing Missing Values (Mean)
# -----
df['Age'].fillna(df['Age'].mean(), inplace=True)
df['Salary'].fillna(df['Salary'].mean(), inplace=True)
df['Experience'].fillna(df['Experience'].mean(), inplace=True)

print("\nDataset After Fixing Missing Values:")
print(df)

# -----
# 3. Outlier Analysis using IQR
# -----
Q1 = df.quantile(0.25)
Q3 = df.quantile(0.75)
IQR = Q3 - Q1

outliers = (df < (Q1 - 1.5 * IQR)) | (df > (Q3 + 1.5 * IQR))

print("\nOutliers Detected (True = Outlier):")
print(outliers)

# Remove outliers
df_cleaned = df[~outliers.any(axis=1)]

print("\nDataset After Removing Outliers:")
print(df_cleaned)

```